STATEMENT OF PURPOSE

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My research objective is to develop **reliable** and **effective** deep learning systems. To build deep learning systems that we can trust and deploy widely, I pursue both **mechanistic understanding** of how they work and **verifiable guarantees** to control their behavior. We need mechanistic understanding to reveal how neural networks learn and adapt, allowing us to better control their training outcomes. We need verifiable guarantees so that these systems reliably follow human intent in complex tasks. Equally important is developing effective and efficient approaches to improve how these systems handle complex sequential decisions. Towards realizing this vision, I have investigated **neural mechanisms** underlying model fine-tuning [1], developed **formal verification** approaches to ensure reliability [2], and designed **structured learning** methods for complex sequential tasks [3, 4, 5].

Understanding and Predicting Cross-Lingual Generalization

Language models demonstrate remarkable ability to generalize across languages, yet we lack a fundamental understanding of the neural mechanisms behind this generalization. Motivated by this challenge, my work [1] **(presented at EMNLP 2024)** investigates the **neural mechanisms that enable cross-lingual generalization** through a case study in toxicity mitigation.

I discovered that safety training in English alone can effectively reduce toxicity across different languages. This work represents the **first demonstration of zero-shot cross-lingual generalization in safety training**, which is particularly crucial since collecting high-quality safety training data in many languages remains a significant technical and resource barrier. Through mechanistic causal analysis, I uncovered that toxic content activates consistent neural patterns across multiple languages. When the model learns to suppress these patterns through English training, the suppression naturally extends to other languages due to the multilingual patterns in neural activations. This finding provides a concrete explanation for how models achieve cross-lingual transfer through shared representational structures.

Building on these mechanistic insights, I developed a method to predict the effectiveness of cross-lingual transfer by analyzing the similarity between sentence representations in different languages. For instance, the accuracy at which we can match English sentences to their German translations based on embedding similarity strongly correlates with how effectively safety training will transfer between these two languages. This capability enables us to **predict cross-lingual generalization before training**, helping assess whether downstream training will be effective for different languages, while also opening new directions for **automated assessment of transfer learning.**

Verifying Language Models for Reliable Plan Generation

Planning is a fundamental capability we seek in AI systems, and language models offer a promising interface through natural language interaction—yet their tendency to hallucinate makes directly generating executable plans unreliable. One promising approach involves equipping language models with a traditional symbolic planner: the language model serves as an interface for parsing instructions and provides high-level oversight in the form of Planning Domain Definition Language (PDDL), while the planner handles the planning process with built-in automation.

However, previous works that evaluate these systems disregard semantic correctness—they consider generations solved by any plan as correct, without the need to align with the instruction. As a response, in my work [2] **(submitted to NAACL 2025**[1]**)**, I proposed a method that **formally verifies the semantic correctness of the generated specifications.** It transforms the generated PDDL specifications into scene graphs, computes an expansion of the goal states, and then performs isomorphism checks with the ground truth PDDL. This method ensures two specifications match if and only if they represent the same underlying planning problem.

Further, we leverage this method to craft a benchmark containing more than 120k PDDL specifications with language descriptions for multiple domains like Gripper and Floor Tile. Surprisingly, while GPT-4o demonstrated an impressive 82% accuracy in generating syntactically valid PDDLs, only 30% of these specifications truly aligned with the original problem descriptions. This stark discrepancy highlights the importance of our benchmark and helps calibrate expectations of what current AI systems can actually achieve. Looking ahead, incorporating such verification approaches directly into systems could enable **high-confidence guarantees of correctness** and develop systems with **self-verifying capabilities**.

[1] Paper under review received positive evaluation (scores: 3.5, 3.5, 4.0/5.0, meta review: 4.0/5.0)

Structured Learning for Complex Sequential Tasks

Complex sequential decision-making remains challenging for deep learning systems, as they struggle to organize and reuse learned behaviors effectively. To address this challenge, my research also investigates **structured learning approaches** through different forms of abstraction and prior knowledge in both motion and task planning.

In my work **[3] (presented at RSS 2021 workshop)**, I developed an augmented option-critic framework that makes learning more efficient by incorporating inductive biases as mathematical constraints directly into the optimization objective, enabling more effective exploration in robotics tasks. Building on this theme of efficient learning, I introduced an in-context world model translation approach **[4] (presented at ICML 2023)** that enables robots to solve long-horizon manipulation tasks by learning to translate between simplified and realistic environment dynamics, enabling one-shot generalization to novel tasks. More recently, I developed a more flexible hierarchical reinforcement learning framework that automatically structures exploration using Linear Temporal Logic specifications, **[5] (presented at CoRL 2024 workshop)**, improving upon traditional subgoal-based approaches by allowing more expressive goal decomposition.

This line of work has led to more effective systems through better exploration strategies and abstractions. Some of these approaches, particularly our work on formal languages, open exciting directions for integration with language models, building on my verification framework [1] to develop more reliable and efficient sequential decision-making systems.

Why School

I am excited to work with Prof. XX and many others to XX. With my diverse background and research experience, I would both thrive in and contribute significantly to XX

References

[1] **Xiaochen Li**, Zheng-Xin Yong, and Stephen H. Bach. "Preference Tuning For Toxicity Mitigation Generalizes Across Languages." accepted to Findings of Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2024

[2] Max Zuo, Francisco Piedrahita Velez, **Xiaochen Li**, Michael L. Littman, and Stephen H. Bach. "Planetarium: A Rigorous Benchmark for Translating Text to Structured Planning Languages." in submission to The Nations of the Americas Chapter of the Association for Computational Linguistics (**NAACL**), 2025.

[3] Zhiao Huang, **Xiaochen Li**, Hao Su. "Efficient Hierarchical Navigation and Manipulation by Constraint-induced Option-Reward Design" accepted to Workshop on Declarative and Neurosymbolic Representations in Robot Learning and Control of Robotics: Science and Systems (**DNR-ROB at RSS**), 2021

[4] Stone Tao, **Xiaochen Li**, Tongzhou Mu, Zhiao Huang, Yuzhe Qin, and Hao Su. "Abstract-to-Executable Trajectory Translation for One-shot Task Generalization." accpeted to International Conference on Machine Learning (**ICML**), 2023

[5] Yichen Wei, **Xiaochen Li**, Jason Xinyu Liu, Naman Shah, Benedict Quartey, Stefanie Tellex, George Konidaris, Akhil Bagaria, "Structured Exploration in Reinforcement Learning by Hypothesizing Linear Temporal Logic Formulas," accepted to Workshop on Learning Effective Abstractions for Planning of Conference on Robot Learning (**LEAP at CoRL**), 2024